

Problem A. Trapezoid Sum

Time Limit 2000 ms

Problem Statement

We have a blackboard with nothing written on it. Takahashi will do N operations to write integers on it.

In the i -th operation, he will write each integer from A_i through B_i once, for a total of $B_i - A_i + 1$ integers.

Find the sum of the integers written on the blackboard after the N operations.

Constraints

- All values in input are integers.
- $1 \leq N \leq 10^5$
- $1 \leq A_i \leq B_i \leq 10^6$

Input

Input is given from Standard Input in the following format:

```
 $N$   
 $A_1$   $B_1$   
 $\vdots$   
 $A_N$   $B_N$ 
```

Output

Print the sum of the integers written on the blackboard after the N operations.

Sample 1

Input	Output
2 1 3 3 5	18

In the 1-st operation, he will write 1, 2, and 3 on the blackboard.

In the 2-nd operation, he will write 3, 4, and 5 on the blackboard.

The sum of the integers written is $1 + 2 + 3 + 3 + 4 + 5 = 18$.

Sample 2

Input	Output
3 11 13 17 47 359 44683	998244353

Sample 3

Input	Output
1 1 1000000	500000500000

Problem B. Building Roads

Time Limit 1000 ms

Mem Limit 524288 kB

Byteland has n cities, and m roads between them. The goal is to construct new roads so that there is a route between any two cities.

Your task is to find out the minimum number of roads required, and also determine which roads should be built.

Input

The first input line has two integers n and m : the number of cities and roads. The cities are numbered $1, 2, \dots, n$.

After that, there are m lines describing the roads. Each line has two integers a and b : there is a road between those cities.

A road always connects two different cities, and there is at most one road between any two cities.

Output

First print an integer k : the number of required roads.

Then, print k lines that describe the new roads. You can print any valid solution.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
4 2 1 2 3 4	1 2 3

Problem C. Wizard in Maze

Time Limit 2000 ms

Problem Statement

A maze is composed of a grid of $H \times W$ squares - H vertical, W horizontal.

The square at the i -th row from the top and the j -th column from the left - (i, j) - is a wall if S_{ij} is # and a road if S_{ij} is ..

There is a magician in (C_h, C_w) . He can do the following two kinds of moves:

- Move A: Walk to a road square that is vertically or horizontally adjacent to the square he is currently in.
- Move B: Use magic to warp himself to a road square in the 5×5 area centered at the square he is currently in.

In either case, he cannot go out of the maze.

At least how many times does he need to use the magic to reach (D_h, D_w) ?

Constraints

- $1 \leq H, W \leq 10^3$
- $1 \leq C_h, D_h \leq H$
- $1 \leq C_w, D_w \leq W$
- S_{ij} is # or ..
- $S_{C_h C_w}$ and $S_{D_h D_w}$ are ..
- $(C_h, C_w) \neq (D_h, D_w)$

Input

Input is given from Standard Input in the following format:

```
H W
C_h C_w
D_h D_w
S11 ... S1W
```

Output

Print the minimum number of times the magician needs to use the magic. If he cannot reach (D_h, D_w) , print **-1** instead.

Sample 1

Input	Output
4 4 1 1 4 4 ..#. ..#. .#.. .#.. .#.. .#..	1

For example, by walking to $(2, 2)$ and then using the magic to travel to $(4, 4)$, just one use of magic is enough.

Note that he cannot walk diagonally.

Sample 2

Input	Output
4 4 1 4 4 1 .##. #### #### .##.	- 1

He cannot move from there.

Sample 3

Input	Output
4 4 2 2 3 3	0

No use of magic is needed.

Sample 4

Input	Output
4 5 1 2 2 5 #.### ####. #..## #..##	2

Problem D. Round Trip

Time Limit 1000 ms

Mem Limit 524288 kB

Byteland has n cities and m roads between them. Your task is to design a round trip that begins in a city, goes through two or more other cities, and finally returns to the starting city. Every intermediate city on the route has to be distinct.

Input

The first input line has two integers n and m : the number of cities and roads. The cities are numbered $1, 2, \dots, n$.

Then, there are m lines describing the roads. Each line has two integers a and b : there is a road between those cities.

Every road is between two different cities, and there is at most one road between any two cities.

Output

First print an integer k : the number of cities on the route. Then print k cities in the order they will be visited. You can print any valid solution.

If there are no solutions, print "IMPOSSIBLE".

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 6 1 3 1 2 5 3 1 5 2 4 4 5	4 3 5 1 3

Problem E. Cycle

Time Limit 2000 ms

Problem Statement

There is a simple directed graph with N vertices numbered from 1 to N and M edges. The i -th edge ($1 \leq i \leq M$) is a directed edge from vertex a_i to vertex b_i .

Determine whether there exists a cycle that contains vertex 1, and if it exists, find the minimum number of edges among such cycles.

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $1 \leq M \leq \min\left(\frac{N(N-1)}{2}, 2 \times 10^5\right)$
- $1 \leq a_i \leq N$
- $1 \leq b_i \leq N$
- $a_i \neq b_i$
- $(a_i, b_i) \neq (a_j, b_j)$ and $(a_i, b_i) \neq (b_j, a_j)$, if $i \neq j$.
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N M
a1 b1
a2 b2
⋮
aM bM
```

Output

If there exists a cycle that contains vertex 1, print the minimum number of edges among such cycles. Otherwise, print **-1**.

Sample 1

Input	Output
3 3 1 2 2 3 3 1	3

Vertex 1 $\rightarrow$ vertex 2 $\rightarrow$ vertex 3 $\rightarrow$ vertex 1 is a cycle with three edges, and this is the only cycle that contains vertex 1.

Sample 2

Input	Output
3 2 1 2 2 3	-1

Sample 3

Input	Output
6 9 6 1 1 5 2 6 2 1 3 6 4 2 6 4 3 5 5 4	4

Problem F. Building Teams

Time Limit 1000 ms

Mem Limit 524288 kB

There are n pupils in Uolevi's class, and m friendships between them. Your task is to divide the pupils into two teams in such a way that no two pupils in a team are friends. You can freely choose the sizes of the teams.

Input

The first input line has two integers n and m : the number of pupils and friendships. The pupils are numbered $1, 2, \dots, n$.

Then, there are m lines describing the friendships. Each line has two integers a and b : pupils a and b are friends.

Every friendship is between two different pupils. You can assume that there is at most one friendship between any two pupils.

Output

Print an example of how to build the teams. For each pupil, print "1" or "2" depending on to which team the pupil will be assigned. You can print any valid team.

If there are no solutions, print "IMPOSSIBLE".

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 3 1 2 1 3 4 5	1 2 2 1 2

Problem G. Towers

Time Limit 1000 ms

Mem Limit 524288 kB

You are given n cubes in a certain order, and your task is to build towers using them. Whenever two cubes are one on top of the other, the upper cube must be smaller than the lower cube.

You must process the cubes in the given order. You can always either place the cube on top of an existing tower, or begin a new tower. What is the minimum possible number of towers?

Input

The first input line contains an integer n : the number of cubes.

The next line contains n integers $k_1, k_2, \dots, k_n$: the sizes of the cubes.

Output

Print one integer: the minimum number of towers.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq k_i \leq 10^9$

Example

Input	Output
5 3 8 2 1 5	2

Problem H. Star or Not

Time Limit 2000 ms

Problem Statement

You are given a tree with N vertices and $N - 1$ edges.

The vertices are numbered $1, 2, \dots, N$. The i -th edge connects Vertex a_i and Vertex b_i .

Determine whether this tree is a star.

Here, a star is a tree where there is a vertex directly connected to all other vertices.

Notes

For the definition of a tree, see [Tree \(graph theory\) - Wikipedia](#).

Constraints

- $3 \leq N \leq 10^5$
- $1 \leq a_i < b_i \leq N$
- The given graph is a tree.

Input

Input is given from Standard Input in the following format:

```
N
a1 b1
⋮
aN-1 bN-1
```

Output

If the given graph is a star, print **Yes**; otherwise, print **No**.

Sample 1

Input	Output
5 1 4 2 4 3 4 4 5	Yes

The given graph is a star.

Sample 2

Input	Output
4 2 4 1 4 2 3	No

The given graph is not a star.

Sample 3

Input	Output
10 9 10 3 10 4 10 8 10 1 10 2 10 7 10 6 10 5 10	Yes

Problem I. Message Route

Time Limit 1000 ms

Mem Limit 524288 kB

Syrjälä's network has n computers and m connections. Your task is to find out if Uolevi can send a message to Maija, and if it is possible, what is the minimum number of computers on such a route.

Input

The first input line has two integers n and m : the number of computers and connections. The computers are numbered $1, 2, \dots, n$. Uolevi's computer is 1 and Maija's computer is n .

Then, there are m lines describing the connections. Each line has two integers a and b : there is a connection between those computers.

Every connection is between two different computers, and there is at most one connection between any two computers.

Output

If it is possible to send a message, first print k : the minimum number of computers on a valid route. After this, print an example of such a route. You can print any valid solution.

If there are no routes, print "IMPOSSIBLE".

Constraints

- $2 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 5 1 2 1 3 1 4 2 3 5 4	3 1 4 5

There is a town with N citizens. It is known that some pairs of people are friends. According to the famous saying that “The friends of my friends are my friends, too” it follows that if A and B are friends and B and C are friends then A and C are friends, too.

Your task is to count how many people there are in the largest group of friends.

Input

Input consists of several datasets. The first line of the input consists of a line with the number of test cases to follow.

The first line of each dataset contains the numbers N and M , where N is the number of town's citizens ($1 \leq N \leq 30000$) and M is the number of pairs of people ($0 \leq M \leq 500000$), which are known to be friends. Each of the following M lines consists of two integers A and B ($1 \leq A \leq N$, $1 \leq B \leq N$, $A \neq B$) which describe that A and B are friends. There could be repetitions among the given pairs.

Output

The output for each test case should contain (on a line by itself) one number denoting how many people there are in the largest group of friends on a line by itself.

Sample Input

```
2
3 2
1 2
2 3
10 12
1 2
3 1
3 4
5 4
3 5
4 6
5 2
2 1
7 1
1 2
9 10
8 9
```

Sample Output

```
3
7
```